UAV-Borne Quasi-3-D Observation of Physical–Biological Coupling in a Dipole Eddy by Integrating Radar Altimetry and Ocean LiDAR

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***Abstract*—3-D remote sensing of the upper ocean remains a major challenge in ocean observation. Conventional dynamic remote sensing can effectively detect sea surface height (SSH), but it lacks the capability to synchronously sense subsurface structures, which limits observations of 3-D physical and biogeochemical variability in the upper ocean. To address this limitation, we develop an uncrewed aerial vehicle (UAV)-borne quasi-3-D remote sensing system that integrates a dual-frequency Ka/Ku radar altimeter and a dual-wavelength blue–green ocean light detection and ranging (LiDAR) system. A flight campaign of over 5800 km was carried out over a mesoscale dipole eddy in the northern South China Sea (SCS), enabling synchronous high-resolution measurements of sea level anomaly (SLA) and subsurface chlorophyll-a (Chl-a) profiles. The SLA retrieved from UAV-borne radar altimetry agreed with the 0.25° Data Unification and Altimeter Combination System (DUACS) and Surface Water and Ocean Topography (SWOT) 2-km products, with root-mean-square errors of 0.44 and 0.51 cm and mean biases of +0.006 and −0.001 cm, respectively. The Chl-a profiles retrieved from UAV-borne LiDAR agreed with synchronous in situ fluorescence observations, with an RMSE of 0.071 $\mu g \cdot L^{-1}$, an MAE of 0.067 $\mu g \cdot L^{-1}$, and a mean bias of +0.067 $\mu g \cdot L^{-1}$. These synchronous observations further revealed the coupled physical and bio-optical structure of the dipole eddy, showing that an approximately 10 cm decrease in SLA coincided with an approximately 12 m shoaling of the subsurface chlorophyll maximum (SCM) and an approximately 0.15 $\mu g \cdot L^{-1}$ increase in subsurface Chl-a. These results support the feasibility of synchronous quasi-3-D sensing of upper ocean dynamical and bio-optical structure and inform the development of future spaceborne 3-D ocean remote sensing missions.**

***Index Terms*—Airborne altimetry, mesoscale eddy, ocean light detection and ranging (LiDAR), physical–biological coupling, uncrewed aerial vehicle (UAV).**

I. Introduction

THREE-DIMENSIONAL remote sensing of the upper ocean remains a major challenge in ocean observation. Existing remote sensing techniques usually characterize either sea surface dynamics or vertically integrated optical signals, whereas direct and synchronous observations of subsurface physical and biological structures are still limited. This observational gap is particularly restrictive in highly dynamic mesoscale systems, where vertical displacement links sea surface signals with subsurface ecosystem variability. A dipole eddy, consisting of an adjacent cyclonic eddy (CE) and anticyclonic eddy (AE), provides a representative setting because the opposing upwelling and downwelling structures can strongly modulate isopycnals, mixed-layer depth, and the subsurface chlorophyll maximum (SCM) within a confined spatial domain [1], [2]. Therefore, characterizing the physical and biological responses within such systems requires synchronous, high-resolution observations of sea level anomaly (SLA) and subsurface biological structure [3]. Yet, bridging this observational gap remains challenging because it requires strict spatiotemporal consistency between surface dynamic signals and subsurface structure observations [4], [5], [6], [7].

Radar altimetry has provided the foundation for observing sea surface dynamics over the global ocean for several decades [8], [9], [10]. Conventional nadir altimeters, including the Jason, Sentinel-3, and HY-2 series, have produced continuous and long-term records of sea surface height (SSH), which are essential for identifying mesoscale eddies and quantify-

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ing large-scale ocean variability [10], [11], [12], [13]. More recently, the Surface Water and Ocean Topography (SWOT) mission has extended altimetric observations from 1-D nadir profiles to wide-swath measurements, substantially improving the spatial sampling of SSH fields and enabling the observation of smaller scale dynamic features [14], [15], [16]. At finer spatial scales, airborne systems such as AirSWOT further demonstrate the potential for high-resolution wide-swath measurements of SSH and provide useful support for satellite validation [17], [18]. In addition, mapped multimission products provide a widely used reference for mesoscale eddy detection and intercomparison [19]. Together, these developments have greatly improved the horizontal observation of ocean dynamics and supported analyses of mesoscale and submesoscale variability in energetic regions such as the South China Sea (SCS) [20], [21], [22], [23], [24]. Despite these advances in spatial resolution [25], radar altimetry still primarily describes surface dynamic expressions and cannot directly resolve the vertical structure of the upper ocean.

Ocean light detection and ranging (LiDAR) systems provide complementary remote sensing capability for probing the ocean interior [26], [27], [28], [29], [30]. Compared with passive ocean color remote sensing, which is mainly sensitive to optically weighted near-surface information, active ocean LiDAR enables the retrieval of depth-resolved optical properties and the inversion of chlorophyll profiles within the water column [30], [31]. Previous studies based on dedicated airborne systems, such as the NASA airborne oceanographic lidar (AOL) and the high spectral resolution lidar (HSRL), together with related developments by the Shanghai Institute of Optics and Fine Mechanics (SIOMs), the Second Institute of Oceanography, Ministry of Natural Resources (SIO/MNR), and Ocean University of China (OUC), have demonstrated the feasibility of detecting subsurface scattering layers and profiling phytoplankton-related structures [27], [28], [32], [33], [34], [35]. These advances demonstrate the capability of ocean LiDAR for high-resolution sensing of the vertical bio-optical structure of the upper ocean.

Recent coordinated applications of SWOT and the Plankton, Aerosol, Cloud, ocean Ecosystem (PACE) mission have enabled the spatiotemporal collocation of wide-swath SSH observations with hyperspectral ocean-color products, thereby improving the characterization of relationships between mesoscale surface dynamics and near-surface biological variability [16], [36], [37], [38], [39]. However, current joint applications primarily combine horizontally resolved sea-surface dynamical fields with optically weighted near-surface bio-optical properties. Further advances are needed in depth-resolved profiling of the ocean water column and in synchronized quasi-3-D observations that link sea-surface dynamics with depth-resolved subsurface physical and bio-optical structure.

To address this observational gap, this study presents an uncrewed aerial vehicle (UAV)-borne quasi-3-D remote sensing system that integrates a dual-frequency Ka/Ku radar altimeter and a dual-wavelength blue–green ocean LiDAR. Together, the synchronously acquired along-track SLA and depth-resolved chlorophyll-a (Chl-a) profiles, complemented by upper ocean thermal profiles reconstructed using a back-propagation neural network (BPNN), constitute a co-registered quasi-3-D section. Here, quasi-3-D refers to a synchronized surface-to-subsurface representation resolved along one horizontal, along-track dimension and the vertical dimension. The objectives of this study are threefold. First, to establish a single-platform synchronous observation approach for sea-surface dynamics and subsurface biological structure. Second, to evaluate the performance of the integrated system and its retrieval products. Third, to characterize the quasi-3-D physical–biological structure of the dipole eddy, with emphasis on how eddy dynamics modulate subsurface chlorophyll structure.

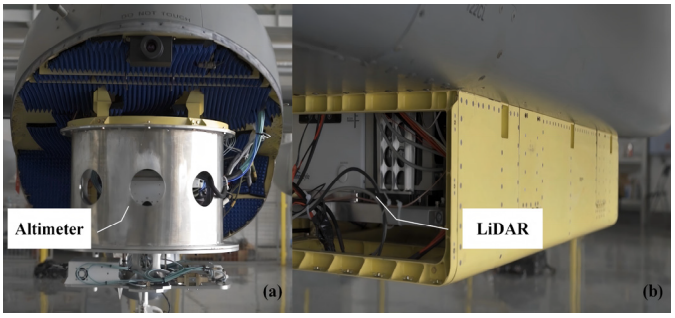


Fig. 1. Core payload subsystems of the UAV-borne quasi-3-D remote sensing system. (a) Dual-frequency Ka/Ku radar altimeter. (b) Dual-wavelength blue–green ocean LiDAR.

II. System and Datasets

A. UAV-Borne Quasi-3-D Remote Sensing System and Synchronous Observation Strategy

Fig. 1 shows the airborne payload configuration, comprising a dual-frequency Ka/Ku radar altimeter and a dual-wavelength blue–green ocean LiDAR integrated on a common UAV platform. The altimeter antenna was installed in the nose section to reduce aerodynamic vibration and maintain a stable nadir-looking geometry, whereas the LiDAR transceiver was housed in the rear fuselage compartment to reduce electromagnetic interference and mechanical vibration. The two sensors share a global navigation satellite system/inertial navigation system (GNSS/INS) and timing system and are synchronized by the GNSS pulse-per-second (PPS) signal, enabling synchronized acquisition and geolocation of the radar and LiDAR observations within a common spatiotemporal reference frame.

The Ku-band SLA sequence was sampled at an average native along-track interval of approximately 19 m. At the mean aircraft speed of 45.9 $m \cdot s^{-1}$, the 100-Hz ocean LiDAR sampling corresponded to an average pulse-to-pulse along-track spacing of approximately 0.46 m. Processed Chl-a profiles were generated at 1-s intervals, corresponding to approximately 46 m along track, whereas the raw water-column profiles had a vertical sampling interval of approximately 0.11 m.

Within this configuration, the radar altimeter performs motion-compensated ranging to obtain SSH measurements, which are subsequently used to derive SLA after reference-surface and geophysical corrections. Meanwhile, the ocean

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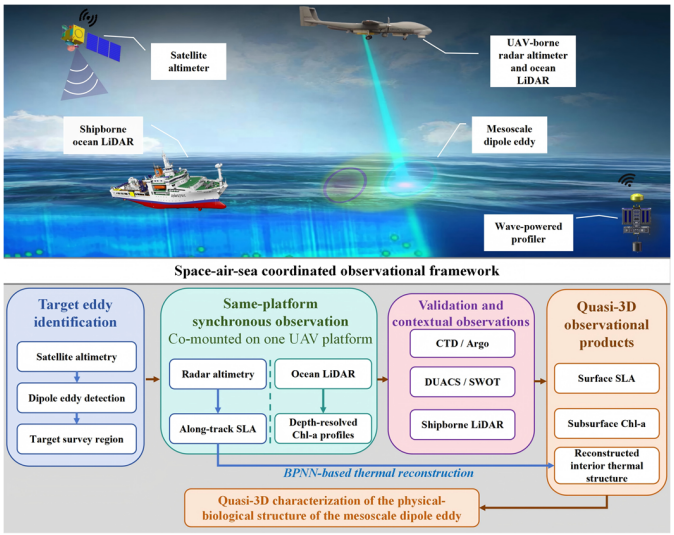


Fig. 2. Collaborative observation strategy and system framework.

TABLE I
TIME WINDOWS OF UAV SORTIES AND IN SITU STATIONS

UAV sorties	UAV time window	UAV duration /distance	Corresponding stations
Sortie 1	6 Apr 2023, 11:30-14:30	3.00 h / 522 km	-
Sortie 2	12 Apr 2023, 06:50-15:40	8.83 h / 1405 km	T4, T5, T7, T9
Sortie 3	20 Apr 2023, 11:20-14:40	3.33 h / 576 km	T11, T13, T15
Sortie 4	21-22 Apr 2023, 21:00-07:00	10.00 h / 1785 km	T17, T19
Sortie 5	25 Apr 2023, 14:20-23:07	8.78 h / 1547 km	H5, H6, H7

Note: All times are reported in local campaign time, UTC+8. The UAV trajectory records were originally time-stamped in UTC and converted to UTC+8. The flight altitude is approximately 3000 m. The shipborne station profiles used in this study were acquired between 12 April 2023, 09:58:55 and 25 April 2023, 21:26:36.

LiDAR acquires water-column backscatter profiles for the inversion of subsurface Chl-a structure. Following retrieval, the radar-derived SLA sequence and LiDAR-derived Chl-a profiles were co-registered with respect to a common cumulative along-track coordinate calculated from the GNSS/INS trajectory. Each 1-s LiDAR profile was matched to the nearest Ku-band SLA sample.

The UAV-borne quasi-3-D remote sensing system was deployed within a hierarchical space–air–sea observing strategy designed for targeted observations of the evolving dipole eddy, as shown in Fig. 2 [40]. The workflow proceeds from large-scale background identification to targeted high-resolution observation. Before the airborne campaign, the region of interest was identified using near-real-time satellite altimetry products. The evolution of the SLA field was analyzed to determine the location, structure, and propagation of the dipole eddy and to guide the observation design. Based on this satellite-derived information, the UAV and a research vessel carried out synchronous transects aligned with the dipole axis. The UAV acquired continuous sections of SLA and subsurface Chl-a, while the research vessel occupied hydrographic stations along the flight track to provide in situ reference profiles. Flight lines and station locations were adjusted before each sortie according to the updated eddy position, in order to reduce spatial mismatch caused by eddy translation and to maintain temporal consistency between the airborne and shipborne observations.

B. Datasets

The primary dataset used in this study consists of UAV observations acquired during the April 2023 field campaign over the dipole eddy system in the northern SCS. Five UAV sorties were completed, with a total flight distance of 5835 km. Table I summarizes the acquisition windows of the five UAV sorties and the shipborne station observations. All times are reported in local campaign time (UTC+ 8). Shipborne observations were collected at 12 stations (H5–H7 and T4–T19) distributed across the CE and AE sectors of the dipole eddy. The spatial distribution of the UAV flight track and shipborne stations is shown in Fig. 3. The closely collocated fluorescence profiles at Stations H5–H7 were used for quantitative evaluation of the UAV-retrieved Chl-a profiles. The selected T-series shipborne LiDAR and hydrographic observations were used to characterize the broader cross-eddy bio-optical and thermohaline structure. Independent Argo temperature profiles were additionally used to evaluate the reconstructed thermal profiles. The selected temperature profiles were acquired by HM2000-type Argo floats and are hereafter referred to as Argo profiles.

Satellite altimetry products were also used for eddy identification and cross-validation. To characterize the mesoscale background and provide a reference for comparison, this study used the Level-4 near-real-time merged SLA product generated within the Data Unification and Altimeter Combination System (DUACS) framework and distributed by the Copernicus Marine Service [19]. Specifically, the product SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046, provided at daily resolution on a $0.25° \times 0.25°$ grid, was used as the synoptic background field for eddy detection and as a mesoscale reference for evaluating the consistency of the SLA retrieved from UAV-borne radar altimetry.

To assess performance at shorter spatial scales, wide-swath SSH observations from the SWOT mission were also used [16]. The analysis employed the April 22, 2023 overpass, which had strong temporal and spatial overlap with the UAV campaign, together with the Level-3 low-resolution expert SSH grid product (version v1.0.2, Cycle 008, Pass 498) distributed by the CNES AVISO+ platform [16]. This dataset provides an independent 2-D reference for kilometer-scale SSH variability along the overlapping segment. In the validation analysis, DUACS and SWOT were used as independent

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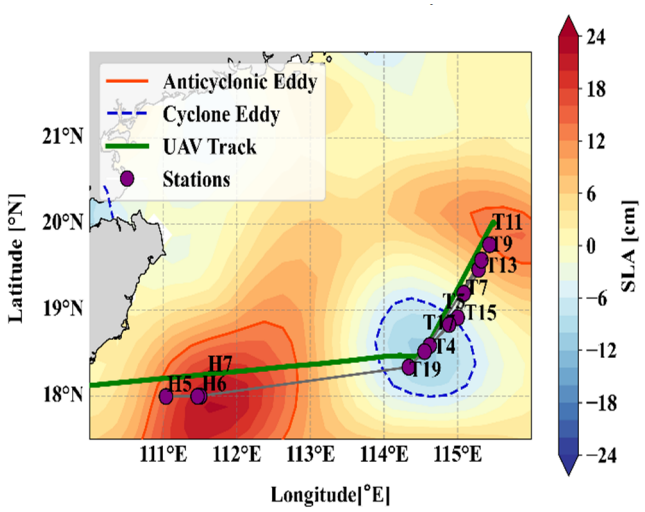


Fig. 3. Map of the experimental campaign over the mesoscale dipole eddy in the SCS. The green line indicates the UAV flight ground track, and the purple circles denote shipborne stations occupied during the coordinated April 2023 field campaign. Stations H5–H7 provided closely collocated fluorescence profiles for quantitative evaluation of the UAV-retrieved Chl-a profiles, whereas the selected T-series stations provided broader bio-optical and hydrographic context.

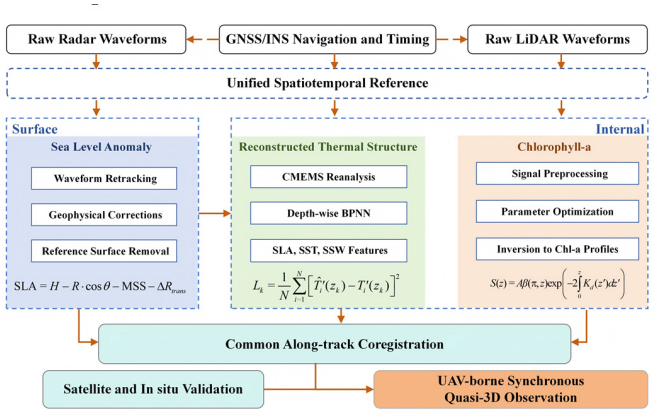


Fig. 4. Processing workflow of the UAV-borne synchronous quasi-3-D observing framework. Raw radar and LiDAR measurements are synchronized and georeferenced using a common GNSS/INS-based spatiotemporal reference. SLA is retrieved through waveform retracking, geophysical corrections, and reference-surface removal. Upper ocean thermal profiles are reconstructed using depth-wise BPNN models trained with CMEMS reanalysis and surface predictors including SLA, SST, SSW, and local SLA-context features. Chl-a profiles are retrieved from LiDAR waveforms through signal preprocessing and PO. The resulting SLA, reconstructed thermal structure, and Chl-a profiles are co-registered on a common along-track coordinate and evaluated using satellite and in situ observations.

external References for evaluating the consistency of the processed UAV-derived SLA after spatial matching.

III. Methodology

To characterize the surface-to-subsurface structure of the mesoscale dipole eddy, this study establishes a synchronized quasi-3-D sectional framework within a common along-track coordinate. The processing workflow is summarized in Fig. 4. UAV-borne radar altimetry retrieves SLA, ocean LiDAR retrieves depth-resolved Chl-a profiles, and upper ocean thermal profiles are reconstructed from SLA and associated surface predictors using the BPNN-based inversion module. The resulting products are jointly analyzed along the common flight-track coordinate to characterize the cross-eddy physical and bio-optical structure.

A. Retrieval of SLA From UAV-Borne Radar Altimetry

The retrieval of SLA from UAV-borne radar altimetry relies on accurate radar ranging and appropriate geophysical corrections. Using the raw dual-frequency Ka/Ku radar echoes and high-frequency GNSS/INS trajectory data acquired by the integrated payload, this study establishes a complete processing chain from signal processing to SLA retrieval.

First, the one-way geometric slant range (R) from the antenna phase center to the instantaneous sea surface is determined through radar pulse compression and waveform retracking. This step converts the hardware sampling timing parameters into physical distance, and the ranging model is expressed as

$$R = \frac{c}{2}\left[t_{start} - \tau_{sys} + \left(k_{retrack} - \frac{N_s}{2}\right)\cdot\Delta\right] \tag{1}$$

where c is the speed of light; t_{start} denotes the sampling start time; τ_{sys} represents the inherent system hardware delay determined through internal calibration; $k_{retrack}$ is the leading edge feature index identified by the retracking algorithm; N_s is the number of range samples (set to 1024 in this system); and $\Delta\tau$ is the time resolution unit after pulse compression.

To evaluate instrumental stability and ranging performance, an internal loop-back calibration mode is used to monitor the equivalent system delay and transmit power. In addition, the absolute ranging performance is assessed over a calm offshore calibration region by comparing the retrieved SSH with the reference mean sea surface (MSS). Subsequently, combining the ellipsoidal height H derived from the high-frequency GNSS/INS trajectory solution with the slant-range estimate, the along-track SSH is first obtained as

$$SSH = H - R\cdot\cos\theta \tag{2}$$

where θ is the antenna off-nadir angle determined from the attitude data and used for slant-range projection onto the local vertical direction. Afterward, the SLA is derived by removing the MSS and applying the required geophysical corrections

$$SLA = SSH - MSS - \Delta R_{trans} \tag{3}$$

where MSS is the reference surface provided by the DTU18 MSS model [41], and ΔR_{trans} denotes the correction term required to place the airborne observations on a consistent oceanographic reference. In this study, ΔR_{trans} mainly includes ocean tide [42], solid Earth tide, pole tide, inverse barometer correction, and residual system bias where applicable. For the low-altitude airborne configuration used here, ionospheric delay is negligible under the aircraft-to-sea-surface propagation geometry, which remains confined to the lower atmosphere. The bulk atmospheric delay below the aircraft is handled through calibration and residual bias treatment. The resulting product represents the UAV-derived SLA after geometric ranging, attitude correction, reference-surface removal,

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and geophysical correction. As the final step of the retrieval processing, along-track filtering was applied to attenuate short-scale fluctuations associated primarily with surface waves and platform vibration, together with residual long-wavelength artifacts related to gradual variations in aircraft altitude.

Because the cross-eddy SLA variations are on the order of centimeters to decimeters, the effective precision of the two radar channels was quantified before the retrieved SLA was used to interpret interior thermal displacement and the associated subsurface bio-optical variability. Accordingly, the calibrated Ku- and Ka-band along-track SSH series were analyzed using the odd–even differencing method described in [43]. The analysis was conducted over continuous track segments to minimize the downward bias that can arise from fitting short data windows.

For a band-specific along-track SSH sequence $X^b[k]$ where $b \in \{\mathrm{Ku}, \mathrm{Ka}\}$, the samples were first partitioned into odd and even subsequences, denoted by $X_o^b[k]$ and $X_e^b[k]$, respectively. If the two subsequences had unequal lengths, the last sample of the longer subsequence is discarded. The odd–even differential sequence is then constructed as

$$X_d^b[k] = X_e^b[k] - X_o^b[k]. \tag{4}$$

To further suppress any residual low-frequency component that may remain after differencing, the sequence $X_d^b[k]$ is linearly fit as

$$X_f^b[k] = a^b k + b^b \tag{5}$$

and the residual sequence is obtained by

$$X_r^b[k] = X_d^b[k] - X_f^b[k]. \tag{6}$$

Let σ_0^b denote the standard deviation of $X_r^b[k]$. Because the odd and even subsequences can be regarded as two independent realizations of the high-frequency measurement error, the effective precision of the original along-track SSH sequence can be written as

$$\sigma^b = \frac{\sigma_0^b}{\sqrt{2}}. \tag{7}$$

The effective 1-Hz precision of each radar channel was estimated using the along-track odd–even differencing procedure defined in (4)–(7). The retrieved SLA was subsequently evaluated against the DUACS and SWOT products in a common along-track reference frame. The gridded reference fields were sampled along the UAV flight path using quadratic spatial interpolation, with cumulative distance calculated from the GNSS/INS trajectory defining the horizontal coordinate. Because the extracted SWOT transect and the radar-altimeter measurements did not share identical sampling locations, the SWOT values were further interpolated to the radar-altimeter positions using 1-D linear interpolation. Statistical metrics were computed only from collocated pairs containing valid values in both datasets, and bias was defined as the reference-product value minus the corresponding radar-altimeter estimate. Because the along-track filtering described above formed part of the retrieval chain, no additional filtering, spatial smoothing, mean removal, or bias correction was applied after collocation.

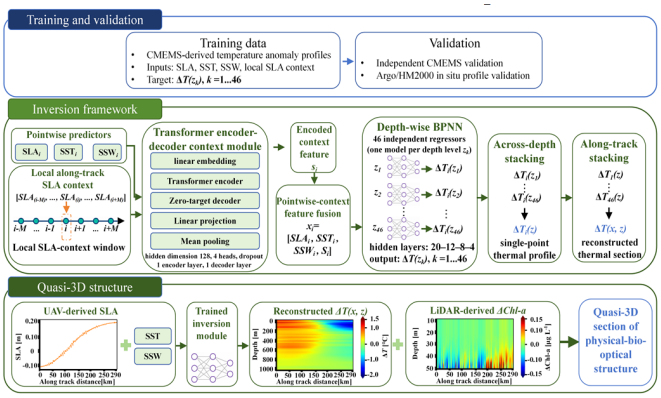


Fig. 5. Architecture and evaluation workflow of the SLA-context-assisted depth-wise BPNN thermal-structure inversion module.

B. Reconstruction of Interior Thermal Structure From SLA by the BPNN-Based Inversion Module

To reconstruct the interior thermal structure along the flight track, the BPNN-based inversion module combines pointwise surface predictors with a local SLA-context feature. The reconstruction complements the directly retrieved radar-altimetry SLA and ocean-LiDAR Chl-a profiles acquired along the same flight track. Its oceanographic rationale follows the established relationships among SSH variability, thermocline displacement, baroclinic adjustment, and vertically integrated density anomalies [44], [45]. Previous satellite–Argo and satellite–reanalysis studies have also shown that SLA, sea surface temperature (SST), sea-surface wind (SSW), and geographic information contain statistical information related to subsurface temperature and thermohaline variability [46], [47]. These relationships motivate the use of the observed surface variables to estimate the subsurface temperature-anomaly structure examined in this study.

The model estimates the nonlinear relationship between the surface predictors and subsurface temperature anomalies [23]. The model was trained using reanalysis data from the Copernicus Marine Environment Monitoring Service (CMEMS) as the reference dataset [48]. At each time step and depth level, the reference temperature was calculated as the mean CMEMS temperature within the spatial domain covered by the UAV track. The supervised temperature-anomaly target was then defined as the local CMEMS temperature minus this contemporaneous, depth-specific reference temperature. The reference temperature was recalculated independently for each time step and depth level. The direct output of each depth-wise model consequently represents a temperature anomaly relative to the contemporaneous mean temperature of the corresponding depth layer.

For each target point, the input comprised pointwise SLA, SST, SSW, and a seven-element local SLA-context feature vector centered on the target location. The local vector summarizes neighboring 1-D SLA variability and provides auxiliary surface information for the inversion. It was processed by a compact transformer-based encoder–decoder feature module. A linear embedding layer mapped the seven-element vector to a 128-D hidden representation, followed by hyperbolic tangent

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activation and dropout regularization. The embedded representation was processed by one transformer encoder layer with four attention heads. The encoder memory and a zero-valued target tensor were then supplied to one transformer decoder layer. Linear projection and mean aggregation produced the encoded SLA-context feature s_i [49]. The encoded SLA-context feature was concatenated with the pointwise SLA, SST, and SSW inputs to form the fused surface-predictor vector. This vector was supplied to a fully connected BPNN containing four hidden layers with 20, 12, 8, and 4 neurons and one linear output neuron. A separate BPNN was trained for each of the 46 discrete depth levels. The direct output of each model is a depth-specific temperature anomaly relative to the contemporaneous mean temperature of the corresponding depth layer. The depth-wise anomaly outputs were assembled across depth to form a temperature-anomaly profile at each target point and then assembled along the UAV track to form the reconstructed temperature-anomaly section. For comparison with Argo absolute-temperature profiles, the corresponding same-time layer-mean temperature was added back to each depth-specific anomaly estimate to recover the absolute-temperature profile. For a sample i and depth z_k, the BPNN estimates the corresponding temperature anomaly as

$$\Delta\hat{T}_i(z_k) = F_k\left(\eta_i, SST_i, SSW_i, s_i; \theta_k\right) \tag{8}$$

where η_i denotes SLA, s_i denotes the encoded local SLA-context feature, and θ_k denotes the trainable parameters of the BPNN at depth z_k. The estimated quantity $\Delta\hat{T}_i(z_k)$ is the temperature anomaly relative to the mean temperature of the corresponding depth layer at the same time. Fig. 5 expands the thermal-reconstruction branch of the overall workflow shown in Fig. 4.

The network parameters were optimized by minimizing the mean squared error between the estimated anomaly and the corresponding CMEMS anomaly defined above

$$L_k = \frac{1}{N}\sum_{i=1}^{N}\left[\Delta\hat{T}_i(z_k) - \Delta T_i(z_k)\right]^2 \tag{9}$$

where N denotes the batch size and k denotes the depth level. The models were optimized using mse loss and Adam with a learning rate of 0.01, a maximum of 1000 epochs, a batch size of 4096, and early stopping with a patience of ten validation checks and a minimum improvement of 1×10^{-5}. Nonoverlapping CMEMS datasets from consecutive years were used for model development and independent evaluation across 46 depth levels from 5 to 1000 m, with the 2014 samples used for training and the 2015 samples reserved for independent evaluation. The trained models were subsequently applied to the April 2023 UAV transect using UAV-derived SLA and collocated SST, SSW, and local SLA-context predictors.

C. Inversion of Subsurface Chl-a Profiles From UAV-Borne Ocean LiDAR

To retrieve synchronous observations of the biological structure of the upper ocean, this study applies a parameter optimization (PO) algorithm to the airborne ocean LiDAR waveforms in order to retrieve depth-resolved Chl-a profiles [35]. In contrast to methods that assume a fixed LiDAR ratio, the PO algorithm iteratively adjusts the relationship between backscattering and attenuation, allowing a better representation of the vertical inhomogeneity of water optical properties. The LiDAR received signal $S(z)$ at depth z follows the LiDAR equation [35]:

$$S(z) = A\beta(\pi, z)\exp\left(-2\int_0^z K_d(z')dz'\right) \tag{10}$$

where A is the system calibration constant, $\beta(\pi, z)$ is the 180° volume scattering function, and K_d is the diffuse attenuation coefficient. Following the classical biological model for Case I waters of Morel and Maritorena [50], which has also been adopted in recent LiDAR-based profile retrieval studies [51], the vertical variation of optical properties is assumed to be primarily controlled by phytoplankton concentration. The dependence of K_d on Chl-a concentration $\langle chl\rangle$ is expressed as

$$K_d = K_w + \chi\langle chl\rangle^e. \tag{11}$$

In the vertical dimension, the effective detection capability of the system is characterized by the maximum penetration depth, defined as the depth at which the signal-to-noise ratio (SNR) of the water column return decreases to unity. Simultaneously, the particulate backscattering component $\beta_p(\pi)$ is parameterized as a power-law function of Chl-a, expressed as

$$\beta_p(\pi) = \frac{b_{bp}}{2\pi\chi(\pi)}, \quad b_p(\lambda) = 0.416\langle chl\rangle^{0.766}\left(\frac{550}{\lambda}\right). \tag{12}$$

Based on these relationships, the inversion algorithm iteratively updates the empirical parameters χ, e, and the phase-function parameter $\chi(\pi)$ within prescribed bounds until the mismatch between the simulated waveform and the measured signal is minimized. Nine consecutive raw range bins were accumulated to form a 1 m depth grid, followed by a 3 m centered vertical smoothing window. The final product is a depth-resolved Chl-a profile along the UAV flight track, from which the depth and concentration of the SCM are further derived for subsequent validation and eddy-scale analysis. The SCM depth and concentration were identified as the depth and magnitude of the maximum Chl-a value within the 10–50 m comparison interval. When the maximum occurred at the lower boundary, it was interpreted as the maximum within the observed depth range rather than as a fully resolved SCM.

To evaluate the Chl-a retrieval results from UAV-borne ocean LiDAR and to provide in situ background information for interpreting the dipole eddy structure, shipborne observations collected during the same field campaign are processed together with the airborne dataset. Among these observations, the fluorescence profiles at Stations H5–H7 were used as independent references for quantitative evaluation of the UAV-retrieved Chl-a profiles. The shipborne ocean-LiDAR profiles and the temperature and salinity observations were used to characterize the broader cross-eddy bio-optical and hydrographic structure.

Shipborne ocean LiDAR Chl-a profiles are retrieved using a regional inversion framework [52]. The raw profiles are preprocessed through sea surface peak detection and alignment, pulse accumulation, background-noise subtraction, and vertical

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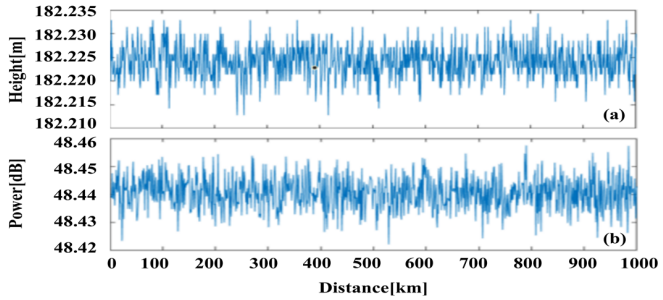


Fig. 6. Internal loop-back calibration of the Ku-band radar altimeter during the field campaign. (a) Equivalent internal delay. (b) Transmit-power monitoring series.

smoothing. A Fernald-type solution of the ocean LiDAR equation is then applied to derive an effective attenuation-coefficient profile, which is converted to Chl-a using a regionally calibrated K_d-Chl relationship based on in situ matchups [52]. Fluorescence measurements are corrected for dark offset and instrument calibration, then converted to Chl-a concentration, processed using the same 3 m centered smoothing window, and median-binned onto a 1 m depth grid to match the effective resolution of the UAV-borne ocean LiDAR retrievals. These processed shipborne biological profiles serve as independent references for evaluating the Chl-a retrieval results from UAV-borne ocean LiDAR.

In parallel, temperature and salinity profiles measured by the multiparameter profiler undergo spike removal, outlier screening, and interpolation onto a uniform depth grid. Based on TEOS-10 [53], potential density anomaly and steric height anomaly are calculated to characterize the cross-eddy thermohaline structure. The steric-height anomaly was used to characterize the dynamical expression of the cross-eddy hydrographic structure and to provide independent physical context for interpreting the SLA retrieved from UAV-borne radar altimetry and the reconstructed temperature-anomaly section.

IV. Results

A. System Calibration and Product Evaluation

Before analyzing the quasi-3-D sectional structure of the dipole eddy, we evaluated the stability of the UAV-borne observing system and the performance of its retrieval and reconstruction products. The evaluation included internal loop-back monitoring of the radar altimeter, offshore calibration, comparisons of UAV-derived SLA with satellite altimetry products, full-profile evaluation of UAV-retrieved Chl-a against closely collocated shipborne fluorescence observations, independent CMEMS evaluation of temperature-anomaly profiles reconstructed using the BPNN-based inversion module, and comparison of reconstructed absolute-temperature profiles with Argo observations.

The internal loop-back calibration records shown in Fig. 6 were used to evaluate the operating stability of the Ku-band radar altimeter during the field campaign. The equivalent internal delay remained centered near 182.225 m throughout the campaign, with only small fluctuations and no evident long-term drift or abrupt jumps, as shown in Fig. 6(a).

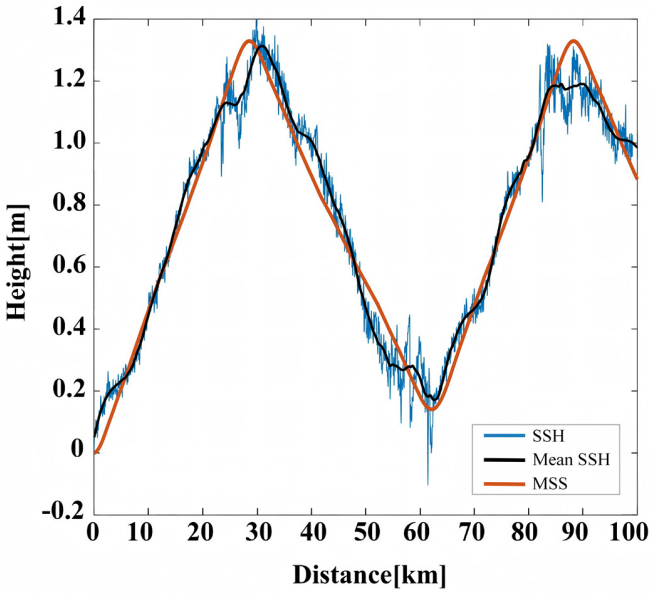


Fig. 7. Comparison of the original UAV-derived SSH, the mean SSH sequence, and the DTU18 MSS reference over an approximately 100 km offshore calibration transect on April 21, 2023. The blue line denotes the original UAV SSH, the black line denotes the mean SSH sequence after smoothing, and the orange line denotes the DTU18 MSS reference.

The corresponding transmit-power series was likewise stable, with peak amplitudes oscillating around the nominal value of 48.44 dB, as shown in Fig. 6(b). Together, these records indicate that the Ku-band radar subsystem remained stable during the airborne observations and provide a consistent basis for subsequent ranging analysis.

As shown in Fig. 7, the offshore calibration result was used to assess the practical ranging performance of the dual-frequency UAV-borne radar altimetry over a geometrically well-constrained offshore transect. The blue line shows the original along-track UAV-derived SSH. The black line shows the same SSH sequence after application of a 1 km centered moving average, corresponding to 53 samples at the average native along-track sampling interval of approximately 19 m. At each position, the smoothed value was calculated as the arithmetic mean of the SSH samples within the centered window. This processing suppresses short-scale measurement fluctuations without introducing an along-track phase displacement and retains the broad-scale SSH variation. The smoothed UAV-derived SSH follows the broad undulation of the DTU18 MSS reference along the calibration transect [41], supporting the broad-scale height consistency of the corrected altimetric measurements. The DTU18 MSS values were not used in calculating the mean SSH sequence.

Using the same calibration-track SSH records, the along-track odd-even differencing analysis described in Section III-A further yields effective 1 Hz precisions of approximately 4.3 cm for the Ku-band channel and 3.5 cm for the Ka-band channel. These complementary assessments characterize the high-frequency precision of the calibrated radar-altimetry sequences and the broad-scale height consistency of the processed SSH, respectively. They support the use

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of the processed UAV-derived SLA for characterizing the cross-eddy sea-level variability examined in the subsequent quasi-3-D physical–biological analysis. For the ocean LiDAR, the effective detection depth reaches about 75 at 486 nm and 60 at 532 nm under a signal-to-noise-ratio threshold of unity, indicating that the validation stations considered here fall within the depth range required to capture the local SCM structure.

The comparisons in Fig. 8 evaluate the ability of the processed UAV-derived SLA to resolve relative sea-level variability associated with the dipole eddy. This capability is first examined against the DUACS Level-4 product over the overlapping segment. As shown in Fig. 8(a) and (b), the SLA retrieved from UAV-borne radar altimetry reproduces the transition from the CE to the AE of the dipole eddy, including both the sign reversal and the amplitude contrast across the section. Over an overlapping transect of approximately 290 km, 15 279 valid pointwise matchup pairs were retained for the DUACS comparison. The comparison yields an RMSE of 0.44 cm, an MAE of 0.328 cm, a mean bias of +0.006 cm, and a pointwise Pearson correlation coefficient of 0.999, indicating that the processed UAV-derived SLA captures the mesoscale geometry of the dipole structure in a manner consistent with the mapped satellite background.

The overlapping segment with SWOT is additionally used to assess the ability of the UAV-borne radar altimetry to resolve relative sea level variability at finer spatial scales. As shown in Fig. 8(c) and (d), the processed UAV-derived SLA and the SWOT Level-3 profile remain closely aligned in both phase and amplitude. The comparison yielded an RMSE of 0.51 cm, an MAE of 0.413 cm, a mean bias of −0.001 cm, and a pointwise Pearson correlation coefficient of 0.666. These results further support the consistency of the processed UAV-derived SLA with independent kilometer-scale satellite observations. Previous validation by the SWOT science team has shown that the Ka-band radar interferometer resolves SSH variability at wavelengths of 20–100 km, with a SWOT–in situ difference corresponding to an RMSE of about 0.56 cm for sub–100 km variability [54]. Against this validation background, the agreement obtained here indicates that the UAV-borne radar altimetry can resolve relative sea level variations of the dipole eddy at spatial scales relevant to mesoscale and submesoscale dynamics. Relative to the satellite product, the denser along-track sampling of the UAV system also retains part of the shorter wavelength variability that appears more muted in the SWOT profile. The RMSE values reported above quantify the spatially matched external consistency of the processed UAV-derived SLA.

The Chl-a retrieval was evaluated using the closely collocated shipborne fluorescence profiles at Stations H5–H7, which overlap with the effective UAV-borne ocean-LiDAR observations. As shown in Fig. 9, the UAV-borne ocean LiDAR captured the depth-dependent Chl-a structure at the validation stations. Across 123 matched depth levels from Stations H5, H6, and H7, the full-profile comparison yielded an RMSE of 0.071 $\mu g \cdot L^{-1}$, an MAE of 0.067 $\mu g \cdot L^{-1}$, a mean bias of +0.067 $\mu g \cdot L^{-1}$, and a correlation coefficient of 0.912, with the bias defined as UAV-borne ocean LiDAR Chl-a minus shipborne in situ fluorescence Chl-a. These statistics were calculated without bias correction. The positive mean bias reflects systematically higher retrieved absolute concentrations relative to the shipborne fluorescence measurements. This offset may arise from differences between active backscatter-based inversion and fluorescence sensing, uncertainty in the

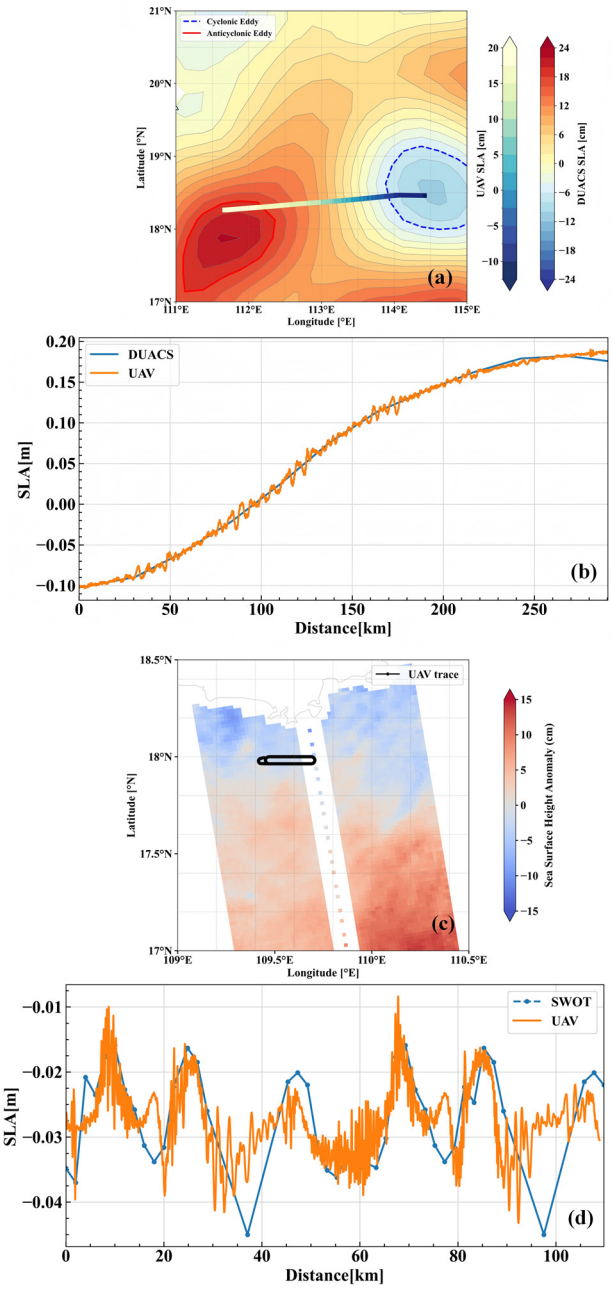


Fig. 8. SLA retrieval results from UAV-borne radar altimetry and cross-validation against satellite altimetry. (a) Flight track overlaid on the DUACS Level-4 gridded SLA field covering the dipole eddy structure. (b) Along-track comparison between the SLA retrieved from UAV-borne radar altimetry (orange) and the DUACS product (blue). (c) Spatial location of the UAV flight segment overlapping with the SWOT Level-3 wide-swath coverage. (d) Fine-scale SLA comparison between UAV observations (orange) and SWOT L3 data (blue) along the overlapping segment. The reported RMSE, MAE, bias, and correlation values quantify the spatially matched external consistency of the processed UAV-derived SLA with the corresponding satellite reference products.

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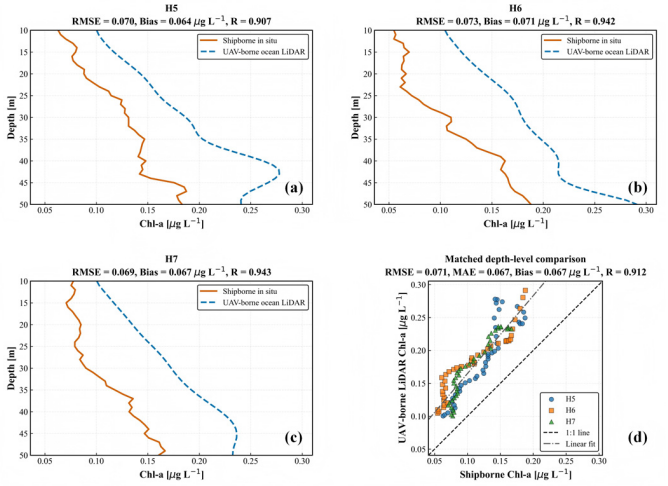


Fig. 9. Comparison of the complete vertical Chl-a profiles retrieved from UAV-borne ocean LiDAR and measured by shipborne in situ fluorescence at Stations H5, H6, and H7. (a) Absolute Chl-a concentration profiles at Station H5 over 10-50 m. (b) Absolute Chl-a concentration profiles at Station H6 over 10-50 m. (c) Absolute Chl-a concentration profiles at Station H7 over 10-50 m. (d) Pooled comparison across 123 matched depth levels from Stations H5, H6, and H7. The black dashed line denotes the one-to-one relationship, and the gray dash-dotted line denotes the linear regression. The statistics were calculated without bias correction.

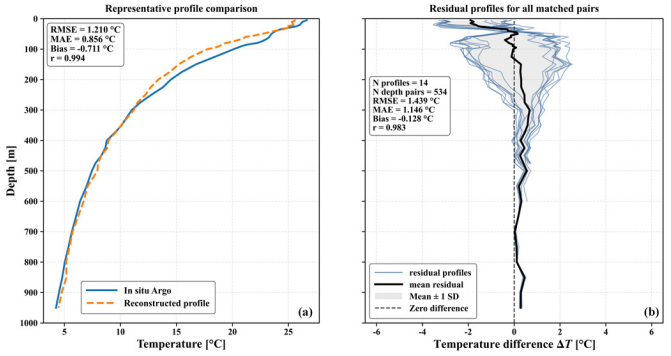


Fig. 10. Validation of absolute-temperature profiles reconstructed by the BPNN-based upper ocean thermal-structure inversion module against in situ Argo observations. (a) Comparison of the objectively selected representative matched pair. The representative pair was selected as the pair whose profile-wise RMSE is closest to the median RMSE across all 14 matched pairs. The blue solid line denotes the in situ Argo profile, and the orange dashed line denotes the reconstructed profile. (b) Residual profiles for all 14 matched pairs. Residuals are defined as reconstructed temperature minus the corresponding in situ Argo temperature. Thin blue lines denote individual residual profiles, the thick black line denotes the depth-dependent mean residual, gray shading denotes the mean ±1 standard deviation, and the vertical dashed line denotes zero residual. No vertical extrapolation was applied.

regional bio-optical relationship, differences in sampling volume, and residual spatial and temporal mismatch between the airborne and shipborne observations.

The temperature-anomaly estimates were evaluated using the independent CMEMS dataset. The evaluation covered 46 depth levels from 5 to 1000 m and included 660 192 valid pointwise pairs. Across all depths, the RMSE, MAE, mean bias, and Pearson correlation coefficient were 1.286 °C, 0.841 °C, −0.147 °C, and 0.501, respectively. Bias was calculated as the temperature-anomaly estimate from the inversion module minus the corresponding CMEMS temperature anomaly. The reconstruction errors generally decreased with depth. The RMSE values were 1.534 °C at 5–100 m, 1.525 °C at 100–300 m, 0.899 °C at 300–600 m, and 0.456 °C at 600–1000 m. Mean biases in the three depth ranges deeper than 100 m were +0.068 °C, −0.117 °C, and −0.067 °C, respectively. The upper 100 m therefore contains the largest reconstruction uncertainty, while the lower RMSE and small mean biases below 300 m indicate greater numerical consistency in the deeper layers.

Absolute-temperature profiles were recovered by adding the contemporaneous mean temperature of each depth layer to the corresponding reconstructed temperature-anomaly estimates. The independent in situ evaluation used 14 Argo profiles located within 10 km of the UAV transect. For each eligible Argo profile, the geographically nearest reconstruction location along the UAV transect was identified, and the corresponding temperature profile reconstructed by the inversion module was extracted. This procedure yielded 14 matched pairs of in situ Argo and reconstructed profiles. Comparisons were performed only at depth levels containing valid observations in both profiles of each pair, producing 534 paired depth-level temperature observations. No vertical extrapolation was applied.

Fig. 10 provides two complementary profile-level assessments of the reconstructed absolute temperature. Fig. 10(a) compares an objectively selected representative matched pair, defined as the pair whose profile-wise RMSE is closest to the median RMSE across all 14 matched pairs. Fig. 10(b) summarizes the residual profiles for all matched pairs. At each valid depth level, the residual is defined as reconstructed temperature minus the corresponding in situ Argo temperature. Across the 534 paired depth-level observations from 5 to 1000 m, the pooled RMSE, MAE, mean bias, and Pearson correlation coefficient were 1.439 °C, 1.146 °C, −0.128 °C, and 0.983, respectively. The residual magnitude and interprofile spread were concentrated mainly within the upper 300 m, where strong mixed-layer and upper thermocline gradients increase sensitivity to local spatial variability and profile-matching differences. The residuals narrowed at greater depths, with RMSE values of 0.569 °C for 300–600 m and 0.265 °C for 600–1000 m. These results indicate that the reconstructed profiles preserve the principal vertical thermal structure sampled by the independent Argo observations. This vertical consistency supports the use of the reconstructed thermal profiles to relate UAV-derived SLA to LiDAR-derived Chl-a structure within the synchronized quasi-3-D section.

Fig. 11 shows negative anomalies and upward-displaced thermal contours in the CE sector, while the AE sector exhibits a downward-displaced thermal structure. Their correspondence with SLA and subsurface Chl-a is examined in Section IV-B.

B. Quasi-3-D Characterization of the Dipole Eddy

Fig. 12 presents the co-located SLA, temperature-anomaly, and Chl-a sections along the cross-eddy UAV transect. Their combined distribution reveals the spatial correspondence

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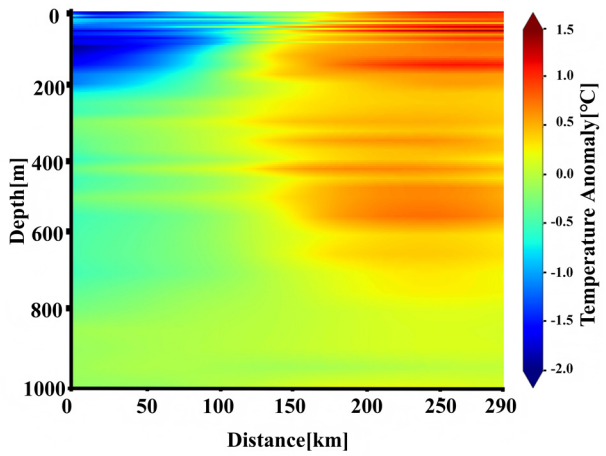


Fig. 11. Cross-eddy temperature-anomaly section constructed from the depth-wise BPNN estimates along the UAV flight track. At each time and depth, the anomaly is defined as the local temperature minus the mean temperature of the same depth layer at that time. The depth-wise estimates were combined along track and interpolated onto a common vertical grid. Positive and negative values indicate warm and cold departures, respectively, from the contemporaneous layer-mean temperature calculated within the spatial domain covered by the UAV track.

between sea-surface dynamics, upper ocean thermal displacement, and subsurface bio-optical structure.

The CE sector is characterized by negative SLA, uplifted upper ocean thermal structure, a shallower SCM, and enhanced subsurface Chl-a. The AE sector is characterized by positive SLA, downward displacement of the upper ocean thermal structure, a deeper SCM, and weaker subsurface Chl-a. The collocated surface and subsurface products therefore reveal coherent spatial co-variability among eddy polarity, SLA, thermal-structure displacement, SCM depth, and subsurface Chl-a distribution along the observed transect.

This correspondence can be expressed more explicitly through the along-track relationship between SLA and SCM depth. An SLA decrease of approximately 10 cm coincides with an SCM shoaling of approximately 12 m and an increase of approximately 0.15 $\mu g \cdot L^{-1}$ in subsurface Chl-a. The strongest horizontal variation in Chl-a occurs in the transition region between the two eddy sectors, where the SLA gradient is also pronounced. This spatial correspondence indicates that the transition region represents a dynamically active part of the observed dipole system, with marked co-variability between sea-surface dynamics and subsurface Chl-a structure.

C. Independent Shipborne Evidence for Cross-Eddy Bio-Optical and Hydrographic Structure

The broader shipborne observations provide independent field evidence for the bio-optical and hydrographic structure of the dipole eddy. As shown in Fig. 13, the shipborne ocean LiDAR-derived Chl-a profiles exhibit clear eddy-polarity-dependent variations. Profiles from the CE sectors, corresponding to Stations T4–T5 and T15–T19 in Fig. 3, generally show shallower SCM, stronger peak concentrations, and sharper vertical gradients. Profiles from the AE sector, corresponding to Stations T9–T13, show deeper SCM and more diffuse subsurface structures. The intereddy station, T7, shows intermediate characteristics. This station-based contrast is consistent with the cross-eddy Chl-a pattern observed by the UAV.

The shipborne hydrographic transect in Fig. 14 places the observed biological contrast within the broader thermohaline structure of the dipole eddy. As shown in Fig. 14(a) and (b), the temperature and salinity sections clearly resolve an alternating cyclonic–anticyclonic–cyclonic pattern, consistent with the spatial organization of the dipole system. In the western and eastern cyclonic sectors, corresponding to Stations T4–T5 and T15–T19, isotherms and isohalines are distinctly uplifted, indicating the upward displacement of the upper ocean interior induced by cyclonic circulation. By contrast, the central anticyclonic sector, corresponding to Stations T9–T13, exhibits a pronounced bowl-shaped depression of the thermohaline structure, characteristic of downwelling. As shown in Fig. 14(c) and (d), the derived potential density anomaly field and steric height anomaly field show the same cross-eddy organization and are consistent with both the SLA retrieved from UAV-borne radar altimetry and the reconstructed thermal section. Taken together, these hydrographic observations provide independent bio-optical and hydrographic evidence that the cross-eddy organization observed along the UAV track is physically consistent with the broader field structure.

V. Discussion and Conclusion

A. Scientific Interpretation and Oceanographic Significance of the Synchronized Quasi-3-D Section

The influence of mesoscale eddies on upper ocean Chl-a distributions has been widely documented using satellite altimetry, passive ocean color, profiling floats, shipborne observations, and numerical models [1], [2], [3], [4], [5], [6], [7], [20], [55]. The scientific contribution of this study lies in resolving coordinated surface-to-subsurface variations across the dipole eddy within a synchronized sectional observation. The co-located SLA, thermal, and Chl-a structures allow the physical and bio-optical expressions of the eddy to be interpreted within a unified spatial and temporal context.

The observed cross-eddy organization is consistent with eddy-induced vertical displacement as an important physical pathway linking SLA variability to SCM shoaling or deepening. In the CE sector, uplift of the upper ocean interior may elevate the nutricline toward the illuminated layer, increasing nutrient availability within the depth range favorable for SCM development, consistent with previous observations in the SCS [55]. The resulting SCM structure is also regulated by the relative positions of the nutricline and the light-defined habitat [56]. Isopycnal stirring, horizontal advection, and frontal processes may further redistribute nutrients, phytoplankton, and chlorophyll-rich water masses near eddy boundaries [1], [57], while Chl-a variability may additionally reflect changes in phytoplankton biomass, cellular pigment content, and photoacclimation [58], [59]. The pronounced SLA and Chl-a gradients in the transition region suggest that vertical displacement and lateral or frontal redistribution may act together in organizing the local bio-optical structure. Enhanced Chl-a should

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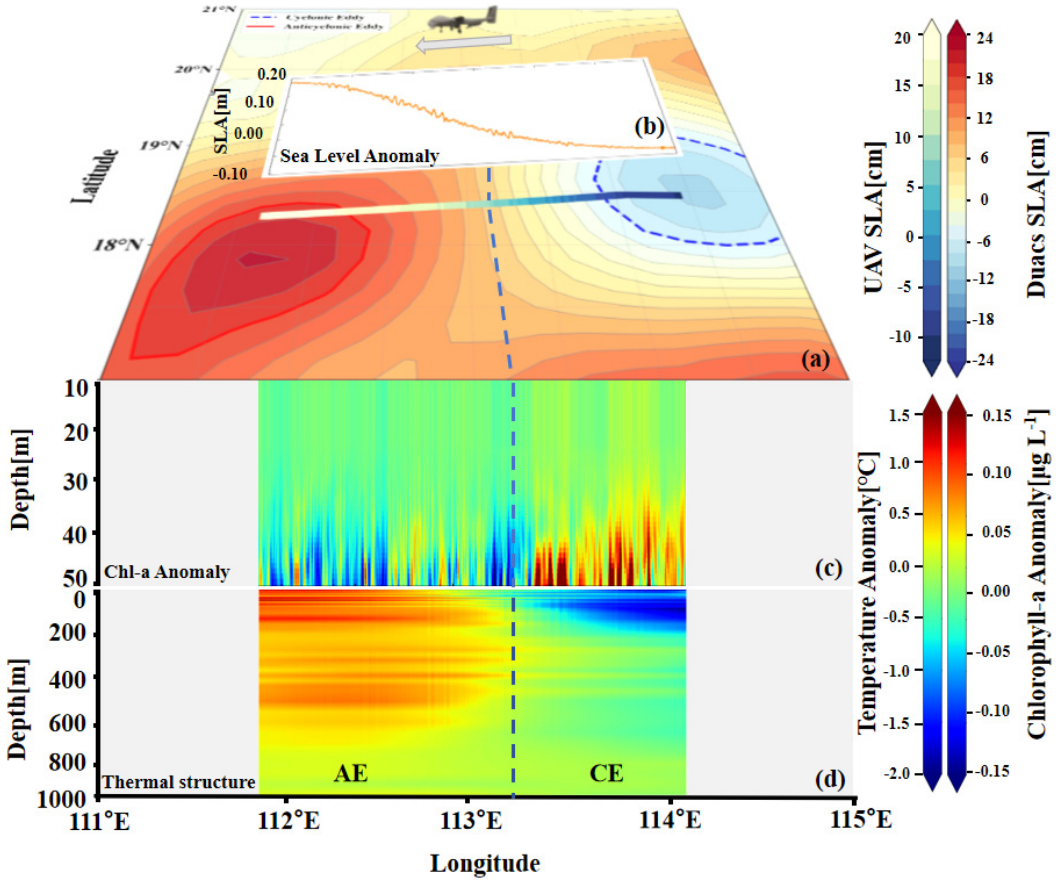


Fig. 12. Quasi-3-D upper ocean physical–biological structure of the dipole eddy. (a) Surface-to-depth sectional visualization of the UAV flight track crossing from the CE to the AE, superimposed on the DUACS SLA background field. (b) Inset showing the along-track SLA comparison between UAV measurements and DUACS data. (c) Vertical section of the subsurface Chl-a anomaly derived from UAV-borne ocean LiDAR. (d) Upper ocean temperature-anomaly section reconstructed by the inversion module.

therefore not be interpreted directly as an equivalent increase in phytoplankton biomass or primary productivity, and the relative contributions of the different physical and biological processes cannot be determined from the present observations alone. The expression of this physical–bio-optical relationship may also vary with eddy evolution. During the generation or intensification stage, vertical displacement and frontal gradients may develop more rapidly than phytoplankton growth and photoacclimation, producing a temporal lag between physical adjustment and the Chl-a response [22], [58], [59]. During the mature stage, coherent uplift in the CE sector and depression in the AE sector may produce clearer polarity-dependent contrasts in SCM depth and subsurface Chl-a [55]. During the decay stage, weakened SLA gradients, lateral stirring, mixing, and lagged biological responses may reduce the surface-to-subsurface correspondence [22], [57], [58], [59]. Interaction between the two eddies of an asymmetric dipole can further modify eddy kinematics and vertical structure [60]. Because the UAV and shipborne observations cover a concentrated campaign window rather than the complete eddy life cycle, the sampled dipole cannot be assigned unambiguously to a specific evolutionary stage. The observed relationship therefore represents the physical and bio-optical state during the campaign period. Repeated observations covering different evolutionary stages, together with measurements of nutrients, underwater light, phytoplankton biomass, community composition, and biological rates, are required to further resolve the underlying processes.

B. Synchronized Observational Capability and Applicability

The methodological contribution of this study is the establishment of a synchronized sectional observing capability that places radar-derived SLA and LiDAR-derived Chl-a profiles within a common spatiotemporal reference. Thermal profiles reconstructed using the independently evaluated BPNN-based inversion module extend this observational context into the upper ocean interior, enabling coordinated interpretation of surface dynamics, thermal displacement, and subsurface bio-optical variability.

Within the broader space–air–sea observing strategy, satellite products provide synoptic background information and target guidance, whereas the UAV system provides targeted and repeatable surface-to-depth sections between satellite overpasses. This complementary capability is particularly relevant to rapidly evolving dynamical processes along mesoscale-eddy margins, including frontogenesis and filament formation, as well as submesoscale eddies. These targets

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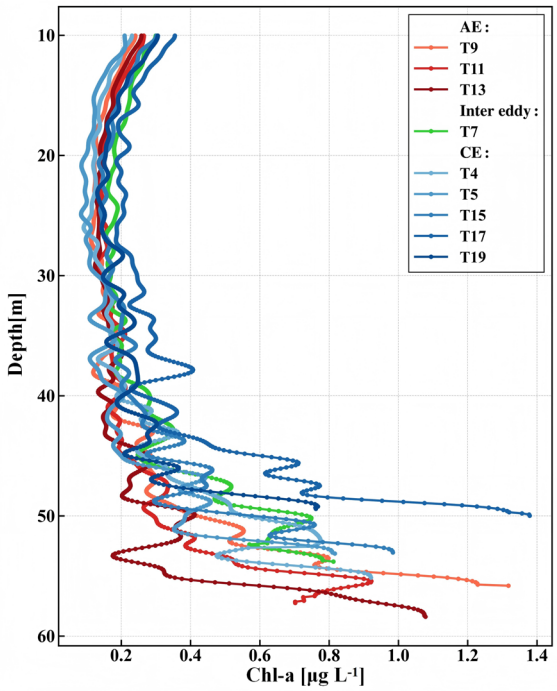


Fig. 13. Shipborne ocean LiDAR-derived Chl-a profiles across the dipole eddy. The profiles are color-coded by dynamical region: AE stations (red hues), CE stations (blue hues), and the intereddy station (green). These profiles provide independent bio-optical evidence for cross-eddy variations in SCM depth and subsurface Chl-a structure.

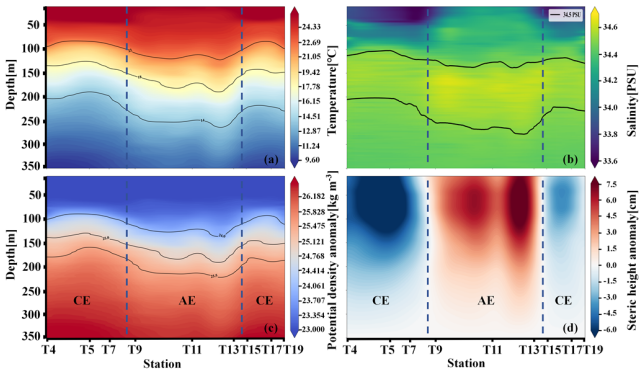


Fig. 14. Shipborne hydrographic sections across the dipole eddy. (a) Temperature. (b) Practical salinity. (c) Potential density anomaly. (d) Steric-height anomaly. The vertical dashed lines indicate the CE–AE sector boundaries.

evolve on spatial and temporal scales shorter than those of the surrounding mesoscale circulation and are often characterized by sharp horizontal gradients and enhanced vertical exchange, making synchronized high-resolution surface-to-depth observations particularly important. Recent wide-swath altimetry observations indicate that upward vertical heat transport associated with submesoscale eddies can be five to ten times larger than that associated with mesoscale eddies [14], while observations of upper ocean convergence show that submesoscale frontogenesis is associated with enhanced vertical velocities and a forward cascade of kinetic energy [61]. Quantifying these rapidly evolving processes requires dedicated repeated-track experiments, for which the synchronized and adaptive quasi-3-D sectional capability demonstrated here provides an observational basis.

The demonstrated along-track quasi-3-D capability provides detailed surface-to-depth information along the flight direction while preserving the spatial correspondence between the physical and bio-optical fields. In the present dipole-eddy application, optimized multileg flight patterns could extend this capability from a single cross-eddy transect to a coordinated set of sections distributed across the eddy system, thereby resolving lateral continuity and spatial variability over a broader area and broadening the quasi-3-D representation of the eddy interior beyond a single transect. Repeated surveys using the same flight lines or multileg configuration could further characterize structural changes during eddy translation and evolution. The achievable spatial extension remains conditioned by variations in ocean-LiDAR penetration and Chl-a retrieval under different water-optical and sea-state conditions, the transferability of the thermal reconstruction, and the horizontal coverage permitted by flight-track geometry and revisit frequency. Continued improvements in sensor performance, adaptive survey design, repeated coverage, and multiplatform coordination can progressively expand the observational domain and support a more complete 3-D characterization of upper ocean structure.

C. Conclusion

This study established and independently evaluated a UAV-borne quasi-3-D observing system that synchronously acquires along-track SLA and depth-resolved Chl-a profiles under a common navigation and timing reference. The BPNN-based inversion module reconstructs complementary upper ocean thermal profiles. Comparisons with satellite altimetry products, shipborne fluorescence observations, CMEMS reanalysis, and Argo profiles support the use of these products for co-registered surface-to-subsurface analysis along a common flight track.

Applied to a mesoscale dipole eddy in the northern SCS, the system resolved coherent cross-eddy variations in eddy polarity, SLA, upper ocean thermal displacement, SCM depth, and subsurface Chl-a. Along the sampled transect, an approximately 10 cm decrease in SLA coincided with an approximately 12 m shoaling of the Chl-a maximum and an approximately 0.15 $\mu g \cdot L^{-1}$ increase in subsurface Chl-a. Together, these co-located variations demonstrate the coordinated expression of dipole-eddy dynamics in upper ocean thermal displacement and subsurface bio-optical structure, providing direct observational constraints for interpreting surface-to-subsurface physical–biological coupling within mesoscale eddies. Future work should focus on repeated observations over the full life cycle of mesoscale eddies and related submesoscale processes. Coordinated deployment of the UAV-borne system with gliders and other autonomous mobile platforms capable of sustained in situ tracking would improve temporal continuity and advance multiscale quasi-3-D remote sensing of rapidly evolving ocean dynamics. Overall, the demonstrated observational capability and field results support the feasibility of synchronous quasi-3-D sensing of upper ocean dynamical

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and bio-optical structure and inform the development of future spaceborne 3-D ocean remote sensing missions.

Author Contributions

Ge Chen, Junwu Tang, and Fangjie Yu conceived the study and provided overall planning and guidance. Songhua Wu and Hongwei Zhang provided methodological guidance on the ocean-LiDAR methodology. Hanwei Sun developed the radar-altimeter payload. Yan He provided the UAV-borne ocean LiDAR system. Fangjie Yu, Ziliang Jiang, Ziming Gao, Mengke Ren, Xinglong Zhang, and Zetao Hu wrote and edited this article.

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